LINEAR ALGEBRA · LESSON 4

Factorization into A = LU

Record elimination **once**, then solve for any new target in two quick passes.

Shown on a real money question: the same funds, many clients.

L from the multipliers

Factor once, solve many

MIT 18.06, Lecture 4

THE PROBLEM

Same funds, new targets every day

Nifty, bond and midcap funds; rules for budget, return and market exposure. The matrix A never changes:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}$$

But each client brings a different target b: ₹10 lakh at 11.1%, ₹10 lakh at 9.8%, ₹15 lakh…

THE IDEA

Don't redo elimination for every client. Save it as $A = LU$.

ONE TOOL FIRST

Undo in reverse order

$$(AB)^{-1} = B^{-1}A^{-1}$$

Check: $AB\,B^{-1}A^{-1} = A\,I\,A^{-1} = I$.

> SHOES AND SOCKS
>
> Socks on, then shoes. To undo: shoes off first, then socks.

Elimination applied E_{21}, E_{31}, E_{32}. To undo it: E_{32}^{-1} first, E_{21}^{-1} last.

From EA = U to A = LU

$$\begin{bmatrix} 1 & 0 \\ -12 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}$$

Move E to the other side (its inverse adds back 12 × row 1):

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}}_{U}$$

Lower triangular with the multiplier; **U**pper triangular with the pivots.

3 × 3

The multipliers go straight into L

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \mathbf{12} & 1 & 0 \\ \mathbf{1} & \mathbf{0.2} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}}_{U}$$

KEY FACT

No row exchanges ⇒ the multipliers $\ell_{21} = 12$, $\ell_{31} = 1$, $\ell_{32} = 0.2$ sit directly in L. No extra work.

WHY NOT E?

E is messy, L is clean

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ \mathbf{1.4} & -0.2 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ \mathbf{1} & 0.2 & 1 \end{bmatrix}$$

Going forward, steps interfere: removing $0.2 \times$ the new row 2 quietly adds back $0.2 \times 12 = 2.4$ of row 1, so $-1 + 2.4 = 1.4$.

Going backward with L, nothing interferes. That's why software stores L and U.

SOLVING

Two triangular passes

Forward: $Lc = b$, top down. **Back:** $Ux = c$, bottom up.

Client: $b = (10, 98, 5.2)$

- $c_1 = 10$, $c_2 = 98 - 120 = -22$, $c_3 = 5.2 - 10 + 4.4 = -0.4$
- $z = 1$, then $-5y + 3 = -22 \Rightarrow y = 5$, then $x = 4$

₹4 lakh Nifty, ₹5 lakh bonds, ₹1 lakh midcap. No new elimination needed.

THE COST

Factor once, solve many



Factoring: about $\frac{1}{3}n^3$ operations. Each new target: about n^2. For 1,000 assets: 333 million once, then 1 million per client.

A VARIATION

A = LDU

Pull the pivots out of U into a diagonal matrix D:

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -0.4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -0.6 \\ 0 & 0 & 1 \end{bmatrix}$$

Now both triangular factors have 1s on the diagonal, and the pivots sit in the middle.

ROW EXCHANGES

PA = LU

A credit fund (12%, like Nifty) puts a 0 in a pivot position. Swap rows first, then factor:

$$P_{23}A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}$$

PERMUTATIONS

$n!$ of them (6 for 3×3). Closed under multiplication. $P^{-1} = P^T$.

WATCH OUT

Singular matrices can have LU too

The balanced fund (half Nifty, half bonds) factors with the **same** L:

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & \mathbf{0} \end{bmatrix}$$

The trouble shows up on U's diagonal: a 0 pivot, so back substitution fails.

> **TEST**
>
> All pivots on U's diagonal non-zero ⇔ invertible.

RECAP

Key takeaways

1. $(AB)^{-1} = B^{-1}A^{-1}$: reverse order.
2. $A = LU$: multipliers in L, pivots in U.
3. L is clean; E is messy.
4. Solve: forward $Lc = b$, back $Ux = c$.
5. $n^3/3$ to factor, n^2 per target.
6. Row swaps: $PA = LU$.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A lab that factors your own matrix into L and U and solves for any target, plus worked hedging and options examples:

learn.krisarj.com

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